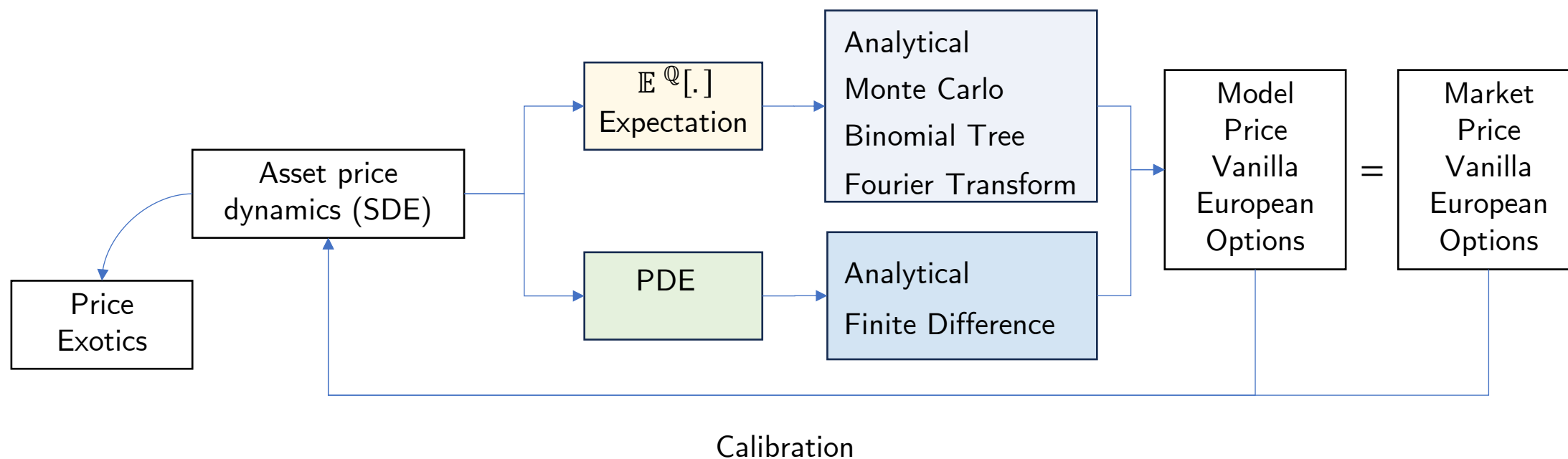


Option Pricing using Fourier Transform

(1. Characteristic functions and Fourier basics)

Pricing approaches

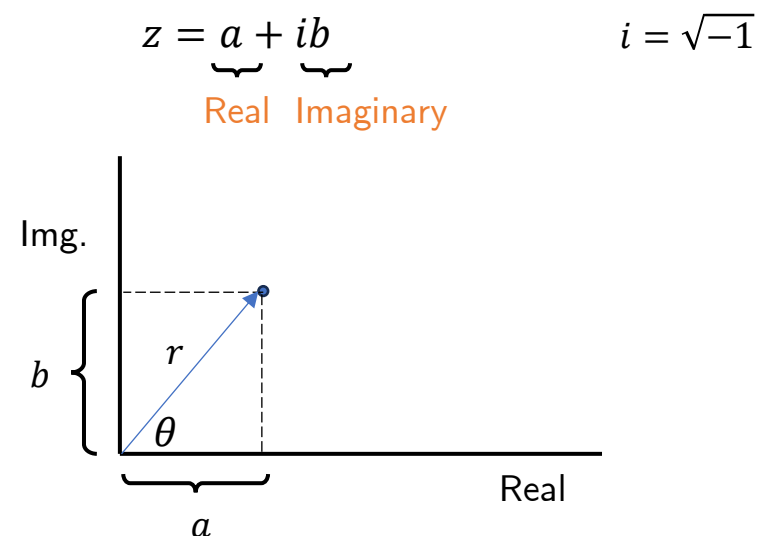


- Analytical solutions are the best (if available)
- Monte Carlo shines in high dimension problems (multiple correlated stochastic processes)
- Binomial Tree and Finite difference are good for low dimensional problems (Some Greeks you get free of cost)
- Fourier Transform provides very fast semi-analytical solution and real time calibration capabilities at scale

Complex Number Recap



- Complex numbers is represented as



- Alternatively, z can be written as $z = re^{i\theta}$ where $r = \sqrt{a^2 + b^2}$ and $\tan \theta = b/a$
- Euler's identity $e^{i\theta} = \cos\theta + i\sin\theta$

Characteristic function and Fourier transform



- The **Fourier transform** and **Inverse Fourier transform** of a function called the **Fourier pair** is defined as

Fourier Transform

$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

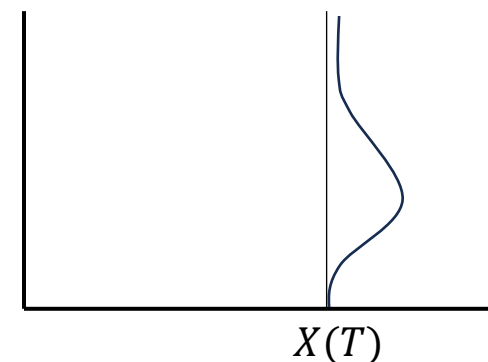
Inverse Fourier Transform

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} F(u) du$$



- The **characteristic function** is the **Fourier transform of the density** function. For European option pricing, we are concerned only with the terminal density $f_{X(T)}(x)$

$$\underbrace{\phi_X(u)}_{\text{Characteristic function}} = \mathbb{E}[e^{iux}] = \int_{-\infty}^{\infty} e^{iux} \underbrace{f_X(x)}_{\text{PDF}} dx$$



Properties of characteristic functions

- Moments can be recovered via differentiation

$$\mathbb{E}[X^k] = \frac{1}{i^k} \frac{\partial^k \phi_X(u)}{\partial u^k} \Big|_{u=0} \quad (\text{Prove it!})$$

- If X and Y are independent, then CF of $X + Y$ will result in product of CFs

$$\phi_{X+Y}(u) = \phi_X(u)\phi_Y(u)$$

- Scaling of a random variable

$$\phi_{aX}(u) = \phi_X(au)$$

Exercise : Show that the characteristic function for a Normal Random $X \sim N(\mu, \sigma^2)$ is given as $\phi_X(u) = e^{iu\mu - \frac{1}{2}\sigma^2 u^2}$

Pricing via Fourier Techniques

- Price of European options depend on the **terminal density** of the stock price. For some cases, the terminal density is available in closed form (Black-Scholes lognormal). But for many other models the density is not available in closed form, however the characteristic function is known. Fourier techniques can price options directly from characteristic function.

Using the convention $\phi_X(u; \tau) = \mathbb{E}_t[e^{iuX_T}]$, $X_T = \ln S_T$, $x = \ln S_t$, $\tau = T - t$

Black-Scholes $\phi_X(u) = \exp\left(iu\left[x + \left(r - q - \frac{1}{2}\sigma^2\right)\tau\right] - \frac{1}{2}\sigma^2u^2\tau\right)$

Bachelier / Normal model Usually written for price/forward F_T , not log-price: $\phi_F(u) = \exp\left(iuF_t - \frac{1}{2}\sigma_N^2u^2\tau\right)$

Merton Jump Diffusion $\phi_X(u) = \exp\left(iu\left[x + \left(r - q - \lambda\kappa - \frac{1}{2}\sigma^2\right)\tau\right] - \frac{1}{2}\sigma^2u^2\tau + \lambda\tau\left(e^{ium_J - \frac{1}{2}\delta_J^2u^2} - 1\right)\right)$ where $\kappa = e^{m_J + \frac{1}{2}\delta_J^2} - 1$

Variance Gamma $\phi_X(u) = \exp(iu[x + (r - q - \omega)\tau])\left(1 - iu\theta v + \frac{1}{2}\sigma^2vu^2\right)^{-\tau/v}$ where $\omega = -\frac{1}{v}\ln\left(1 - \theta v - \frac{1}{2}\sigma^2v\right)$

Heston $\phi_X(u) = \exp(iux + iu(r - q)\tau + C(u, \tau) + D(u, \tau)v_t)$

Bates $\phi_X^{Bates}(u) = \phi_X^{Heston}(u; r - q - \lambda\kappa) \times \exp\left(\lambda\tau\left(e^{ium_J - \frac{1}{2}\delta_J^2u^2} - 1\right)\right)$

Properties of Fourier Transform

In Signal processing, Fourier transform transforms a signal from time domain (t) to frequency domain (ω). Inverse Fourier transforms it back to time domain. For our application, we are using x instead of t and u instead of ω . Here are some the important properties of Fourier transform

$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

Property	Formula
Linearity	$\mathcal{F}\{af_1(x) + bf_2(x)\} = aF_1(u) + bF_2(u)$
Shift in x	$\mathcal{F}\{f(x - x_0)\} = e^{iux_0} F(u)$
Shift in u	$\mathcal{F}\{f(x)e^{iu_0x}\} = F(u + u_0)$
Scaling in x	$\mathcal{F}\{f(ax)\} = \frac{1}{ a } F\left(\frac{u}{a}\right)$
Reversal in x	$\mathcal{F}\{f(-x)\} = F(-u)$
Differentiation in x	$\mathcal{F}\left\{\frac{d}{dx} f(x)\right\} = -iuF(u)$

Common Fourier pairs



The Fourier pair for some commonly used functions are given below. Note that if the function is spread out in x domain, its Fourier transform will be localized(impulse) in u domain and vice versa.

Function : $f(x)$	Fourier Transform : $F(u)$
$\delta(x)$	1
$\delta(x - a)$	e^{iua}
1	$2\pi\delta(u)$
e^{-iax}	$2\pi\delta(u - a)$
$\cos(ax)$	$\pi[\delta(u - a) + \delta(u + a)]$
$\sin(bx)$	$i\pi[\delta(u - b) - \delta(u + b)]$
$e^{-ax}\mathbf{1}_{x \geq 0}, \quad a > 0$	$\frac{1}{a - iu}$

The Dirac Delta function

- Let's get familiar with Fourier transform of some basic functions. To begin with we need to understand the **Dirac Delta** function. It's best understood as infinitely narrow spike with area 1. It's defined as

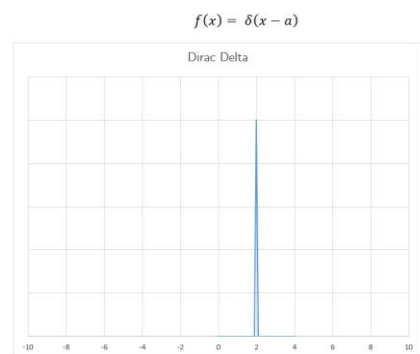
$$\delta(x) = 0, \quad x \neq 0$$
$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

- Shifted delta function $\delta(x - a)$ which will have a spike at $x = a$
- Integrating a function times the delta function just **picks the value of the function where the spike is located**.

$$\int_{-\infty}^{\infty} \delta(x - a) f(x) dx = f(a)$$

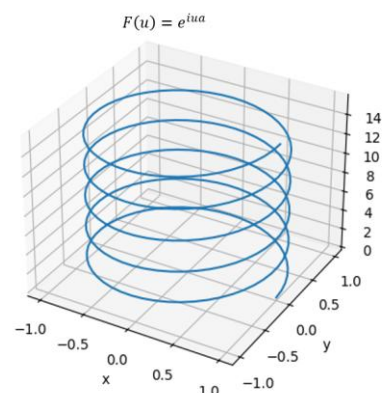
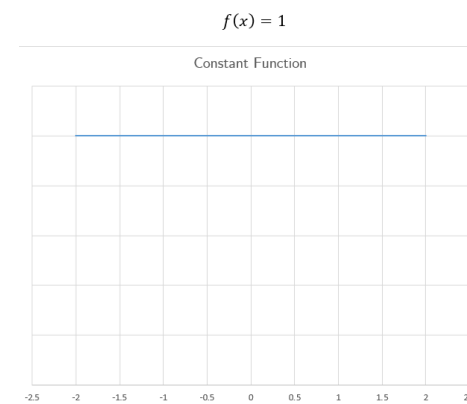
Exercise : Show that the Fourier transform of delta function is 1

Fourier Transform Visuals



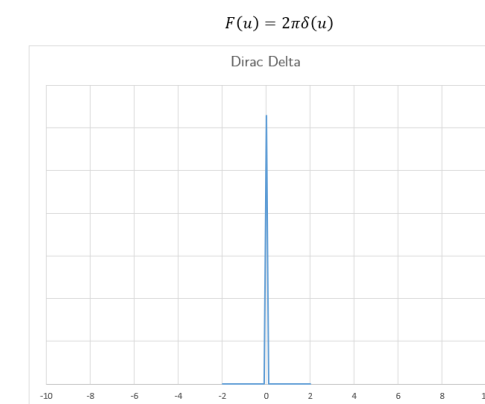
$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} F(u) du$$

 u 

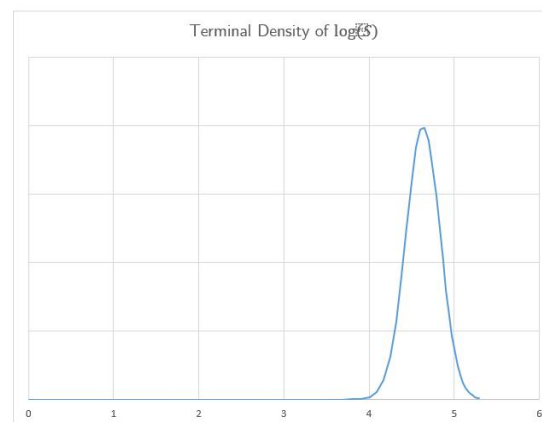
$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} F(u) du$$



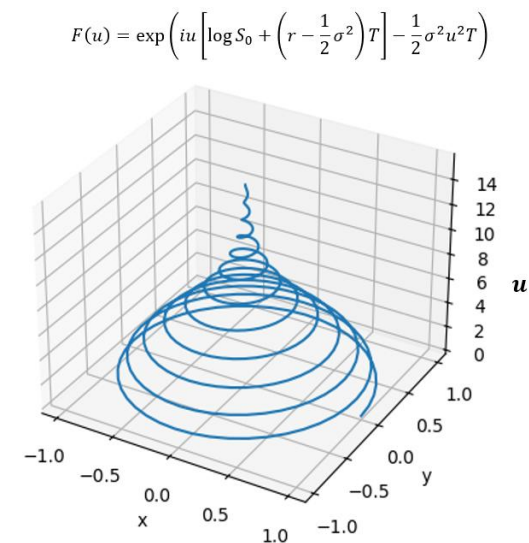
$X = \log(S_T)$

$$X \sim N\left(\log S_0 + \left(r - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right)$$



$$F(u) = \int_{-\infty}^{\infty} e^{iux} f(x) dx$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iux} F(u) du$$

 u